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Submission Date: September 20, 2026

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Research Topic: Evolutionary Biology / Behavioral Science

Title: Are We Cavefish? Environmental Adaptation in an Increasingly Technology-Mediated World

1. Research Problem

Technology is now a core part of our day to day lives in the 21st century. We get information through screens, communicate through platforms, study using digital tools, and spend some of our free time in virtual spaces. This made me wonder, what happens when the environment around an organism changes the information it can receive? Would it change which behaviors are useful as a result?

Cave-dwelling populations of *Astyanax mexicanus* are an interesting biological example of this. In the paper "Regressive evolution in *Astyanax* cavefish," Jeffery describes cave animals as showing the "loss of eyes and pigmentation" and connects these regressive traits with life in permanent darkness [1]. However, cavefish do not just lose traits. Jeffery also describes "constructive traits" that developed alongside regressive ones, including changes in feeding and sensory systems [2]. These include larger jaws, more taste buds, larger cranial neuromasts, and increased fat reserves compared with surface-dwelling fish [2]. This makes cavefish interesting because adaptation is not simply about losing something that is no longer useful. Other traits can become more important in the same environment.

Rodríguez-Morales also describes changes in the lateral-line system of blind cavefish. The lateral line is an important non-visual sensory system, and some cave-dwelling populations show morphological expansion of this system, including changes involving neuromasts and hair cells [3]. This suggests that changes in the environment can affect which sensory systems become more important for obtaining information.

Humans are obviously very different from cavefish. Still, there has been research into how digital technology relates to attention, media multitasking, and online social interaction, especially in children and adolescents [4]–[7]. Cain *et al.*, for example, found that more frequent media multitasking in adolescents was associated with poorer performance on standardized mathematics and English achievement tests and with poorer working-memory performance. However, the associations were not found across every measure they tested. Media multitasking was not related to measures such as processing speed, implicit learning, or manual dexterity [5]. Santos *et al.* also reviewed 11 studies and found that most reported an association between screen time and attention in children, while also noting the need for further research [6].

What interests me is that these two areas are usually researched separately. Cavefish research looks at environmental and sensory adaptation, while human research often looks at technology, attention, and behavior. I want to see what happens if these areas are put next to each other instead.

There is also an obvious problem with making this comparison. A cavefish changing over many generations is not the same as a person developing a habit or adjusting to technology. Shimada,

Ishii, and Shibao distinguish evolutionary-biological adaptation from other kinds of change, including physiological adjustment and phenotypic plasticity [8]. Debevec, Longman, and Bourgois also point out that terms such as response, adaptation, acclimation, acclimatization, and habituation are not interchangeable. They propose using "adaptation" more carefully because these terms can describe different timescales and mechanisms of change [9]. Because of this, this study will examine the analogy rather than assume that humans are biologically evolving to become more like cavefish.

2. Motivation and Cultural Origins

The idea behind this research did not start entirely with biology actually. Humanity has been imagining organisms that are suited to environments very different from ours for a long time, dating back all the way to the 1800s.

In H. G. Wells's *The Time Machine* (1895), humanity becomes divided into the Eloi and Morlocks, with their different environments playing an important role in that division [10]. Popular culture later produced other imagined nonhuman forms. The Betty and Barney Hill case became an important influence on the modern "grey alien" image in American culture [11]. Aleister Crowley's 1919 portrait of "Lam" is another example of an imagined being that was later interpreted by some people through an extraterrestrial lens [12].

None of these examples are scientific evidence for this research. They are more like the cultural background that led me to the question. Humans keep imagining beings that are adapted to worlds that are completely different from our own.

That made me wonder: What happens when humans stop imagining alien environments and start constructing them?

A digital environment is obviously not a cave, and a human is obviously not a cavefish. I am not trying to say they are the same thing. Instead, I want to see whether some of the same ideas can be used to discuss both cases, especially changes in available information, sensory or informational reliance, behavior, and social interaction.

3. Literature Review

Research on *A. mexicanus* shows that adaptation to cave environments involves more than simply losing eyes. Jeffery examines both the loss of eyes and pigmentation and the developmental mechanisms involved in these regressive traits [1], [2]. Cavefish also show constructive features, including larger jaws, more taste buds, larger cranial neuromasts, and increased fat reserves compared with surface fish [2]. The existence of both regressive and constructive traits makes *A. mexicanus* useful for studying how an organism changes as a whole rather than treating adaptation as the loss of one particular characteristic.

Sifuentes-Romero *et al.* examine eye regression as an example of evolutionary trait loss and discuss the developmental, genetic, and evolutionary mechanisms that may contribute to the

repeated reduction of eyes in cavefish [13]. This is important because eye regression is not just an individual fish responding to darkness during its lifetime. It is a repeated evolutionary pattern that has occurred in different cave populations.

Rodríguez-Morales describes both “regressive and constructive traits” and reports that the lateral-line system can show morphological expansion in blind cavefish, providing another way of sensing the surrounding environment when vision is reduced [3]. The lateral line is therefore an example of how the cavefish sensory system is not simply becoming less capable. Instead, different ways of obtaining information can become more important.

Human research deals with a very different kind of environment. Kirjakovski discusses changes in the relationship between perception, cognition, and the information presented by digital environments [15]. Vedechkina and Borgonovi's review also suggests that the relationship between digital technology and attention is not a simple one. Their review emphasizes differences between technologies, users, circumstances of use, and interactions between these factors [4].

Cain *et al.* found that frequent media multitasking in adolescents was associated with poorer academic achievement and working-memory performance, as well as greater impulsivity and lower growth mindset [5]. At the same time, they did not find the same relationship for every cognitive or personality measure. This is important because it suggests that technology-related behavior cannot automatically be treated as having one general effect on every aspect of cognition.

Santos *et al.* conducted a systematic review of research on screen time and attention in children. Their search identified 498 articles, 41 were fully examined, and 11 studies met their inclusion criteria. Most of the included studies reported an association between screen time and attention, although the authors also emphasized the need for further research [6].

Technology also changes how people communicate with each other. Towner *et al.* reviewed 19 quantitative studies involving participants aged 10–22 from countries in North America, Europe, and Asia [7]. Their review examined online and offline self-disclosure between adolescents and peers. Overall, online self-disclosure appeared to be less fulfilling for relationship quality than offline self-disclosure, although some groups seemed to benefit more from online interaction [7].

A newer meta-analysis by McDonald *et al.* provides another example of why the human evidence is not simple. Their 2026 review included 15 articles, with 13 included in the meta-analysis, covering 10,015 participants with a mean age of 16.75 years [14]. They found a small positive association between social-media use and total empathy, while the separate associations with cognitive and affective empathy were not statistically significant [14].

These findings make it difficult to describe technology as simply good or bad for human behavior. The effects seem to depend on what technology is being used, who is using it, what they are doing with it, and the circumstances in which it is being used [4]–[7], [14].

The word “adaptation” also needs to be handled carefully. Shimada, Ishii, and Shibao distinguish evolutionary-biological adaptation from changes that can happen within an individual's lifetime, including phenotypic plasticity and learning [8]. Debevec *et al.* similarly separate terms such as “response,” “adaptation,” “acclimation,” “acclimatization,” and “habituation” [9]. For example, acclimatization occurs within an organism's lifetime, while evolutionary adaptation involves changes across generations. Habituation can also refer to a reduction in response to a repeated stimulus without necessarily involving a deeper biological or functional change [9]. This matters for the present research because a person changing their behavior around technology is not automatically undergoing biological evolution.

4. Research Gap

There is already a lot of research on cavefish adaptation and on human responses to technology. The gap I am interested in is what happens when these two areas are directly compared.

The cavefish literature describes how environmental conditions such as permanent darkness are connected with changes in sensory systems, morphology, feeding, and behavior [1]–[3], [13]. The human literature, meanwhile, examines changes in attention, media multitasking, information processing, self-disclosure, and social functioning in technology-mediated environments [4]–[7], [14], [15]. These studies do not normally treat the two environments as part of the same comparison.

This study will put findings from cavefish research and human technology research into the same categories and then examine how well they actually fit together. The comparison might show a strong parallel, only a partial similarity, or a point where the analogy stops working.

That last result would still be useful. If human behavior does not fit well into a biological adaptation model, then identifying where and why the comparison fails could help show what makes human responses different.

5. Research Questions

Main Research Question

To what extent can environmental, sensory, and behavioral adaptations observed in cave-dwelling *Astyanax mexicanus* provide a framework for understanding human responses to technology-mediated information and social environments?

Sub-Research Questions

RQ1. What environmental, sensory, and behavioral adaptations have been documented in cave-dwelling *A. mexicanus*, and how are they related to changes in available information?

RQ2. What changes in attention, information use, behavior, and social interaction have been documented in humans living in technology-mediated environments?

RQ3. To what extent are the mechanisms and outcomes found in cavefish and humans comparable, and where does the comparison break down?

6. Objectives

1. Identify major environmental, sensory, and behavioral adaptations documented in cave-dwelling *A. mexicanus*.
2. Identify documented changes in human attention, information use, behavior, and social interaction associated with technology-mediated environments.
3. Compare the mechanisms and outcomes found in the two bodies of research.
4. Distinguish evolutionary adaptation from behavioral adaptation, acclimatization, habituation, and learning.
5. Examine both the usefulness and the limits of the cavefish analogy.

7. Methodology

This research will use a systematic comparative literature review rather than direct experimentation on humans or animals. The review will focus on two main groups of studies: research on cave-dwelling *A. mexicanus* and research on human responses to technology-mediated information and social environments.

Relevant studies will be found through academic databases and scholarly search systems. Search terms will include *A. mexicanus*, cave adaptation, sensory systems, behavior, lateral line, digital environment, attention, media multitasking, information processing, adolescent social interaction, and social media.

Studies will be selected based on their relevance to the research questions and the clarity of their methods. Studies will be included when they provide evidence about environmental or technological conditions, sensory or information use, behavior, social interaction, or possible mechanisms connecting these factors.

Information from the selected studies will be organized into common categories. These will include environmental condition, information available, sensory or information reliance, behavioral response, social interaction, proposed mechanism, type of adaptation or adjustment, trade-offs, and limitations.

The analysis will first address RQ1 by identifying environmental, sensory, and behavioral changes reported in cavefish. It will then address RQ2 by organizing findings about human responses to technology-mediated environments. Finally, the two sets of findings will be compared using the same categories to address RQ3.

The comparison will be described as parallel, partially parallel, or different where appropriate. Evolutionary adaptation will be kept separate from behavioral adaptation, acclimatization,

habituation, learning, and technological preference. Studies that contradict the analogy or show its limitations will also be included rather than removed from the analysis.

8. Expected Findings and Significance

The research may find some similarities between cavefish adaptation and human responses to technology-mediated environments. These could include relationships between environmental conditions, information availability, information reliance, and behavior. At the same time, major differences are expected. Cavefish adaptation is evolutionary, while many human responses to technology happen through behavior, cognition, development, or culture.

The existing human evidence is already mixed. Research on attention, media multitasking, self-disclosure, and empathy has found different relationships depending on the technology, individual, behavior, and context [4]–[7], [14]. Because of this, I do not expect the comparison to produce a simple “technology changes humans in this one way” conclusion.

The main point of the study is not to argue that humans are “becoming cavefish.” I want to find out whether the cavefish example is actually useful for thinking about how humans adjust to technological environments.

There is one thing that makes the human case especially interesting. Humans can deliberately build new environments and then change their behavior around the environments they have created. A cavefish cannot design its cave, while humans can design technologies that change what information they see, how they communicate, and what kinds of behavior become normal.

By comparing the similarities and the limits of the cavefish analogy, this research may provide a different way of thinking about how technological environments influence human information use, behavior, and social interaction.

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